

Loam. Well-structured, fertile soil that is moisture-retentive but free-draining.

Marginal water plant. A plant that grows partially submerged in shallow water or in moist soil at the edge of a pond.

Midrib. The main, central vein of a leaf or the central stalk to which the *leaflets* of a *pinnate* leaf are attached.

Monocarpic. Flowering and fruiting only once before dying; such plants may take several years to reach flowering size.

Mulch. A layer of organic matter applied to the soil over or around a plant to conserve moisture, protect the roots from frost, reduce the growth of weeds, and enrich the soil.

Naturalize. To establish and grow as if in the wild.

Nectar. A sweet, sugary liquid secreted by the **nectary**—glandular tissue usually found in the flower but sometimes found on the leaves or stems.

Neutral [of soil]. With a *pH* value of 7, the point at which soil is neither *acidic* nor *alkaline*.

Node. The point on a stem from which a leaf or leaves arise.

Obovate [of leaves]. Egg-shaped in outline, but broadest above the middle; length is 1.5–2 times the width.

Offset. A small plant that arises by natural vegetative reproduction, usually at the base of the mother plant.

Opposite [of leaves]. Borne 2 to each *node*, one opposite the other.

Ovary. The part of the female portion of the flower, containing embryonic seeds, that will eventually form the *fruit*.

Ovate [of leaves]. Egg-shaped in outline, but with a tapering point and broadest below the middle; length is usually 1.5 times the width.

Palm. An evergreen *tree* or *shrublike* plant, normally single-stemmed, with *palmate* or *pinnate* leaves usually in terminal rosettes; strictly a member of the family *Palmae*.

Palmate. Lobed in the fashion of a hand, with 5 lobes arising from the same point.

Pan. A shallow, free-draining pot in which alpine plants or bulbs are grown.

Panicle. A branched *raceme*.

Papilla (pl. *papillae*). A minute protuberance or glandlike structure.

Pealike [of flowers]. Of the same structure as a pea flower.

Peat bed. A specially constructed area, edged with peat blocks and containing moisture-retentive, acidic, peaty soil.

Pedicel. The stalk of an individual flower.

Peduncle. The stalk of a flower cluster.

Peltate [of leaves]. Shield-shaped, with the stalk inserted toward or at the center of the blade and not at the margin.

Perennial. Living for at least 3 seasons. In this book the term when used as a noun, and unless qualified, denotes an *herbaceous* perennial. A woody-based perennial dies down only partially, leaving a woody stem at the base.

Perianth. The outer parts of the flower comprising the *calyx* and the *corolla*. The term is often used when the calyx and the corolla are very similar in form.

Perianth segment. One portion of the *perianth*, resembling a *petal* and sometimes known as a *tepala*.

Petal. One portion of the often showy and colored part of the *corolla*. In some

families, e.g. *Liliaceae*, the *perianth segments* are petallike and referred to horticulturally as petals.

Petaloid. Like a petal.

Petiole. The stalk of a *leaf*.

pH. The scale by which the acidity or alkalinity of soil is measured. See also *acidic*, *alkaline*, *neutral*.

Phyllode. A flattened leaf stalk, which functions as and resembles a leaf.

Pinch out. To remove the growing tips of a plant to induce the production of side-shoots.

Pinna (pl. *pinnae*). The primary division of a *pinnate* leaf. The fertile *pinnae* of ferns produce spores, vegetative *pinnae* do not.

Pinnate [of leaves]. Compound, with *leaflets* arranged on opposite sides of a central stalk.

Pistil. The female part of a flower comprising the *ovary*, *stigma* and *style*.

Pollard [of a tree]. To cut back to its main branches in order to restrict growth.

Pollination. The transfer of pollen from the *anthers* to the *stigma* of the same or different flowers, resulting in fertilization of the embryonic seeds in the *ovary*.

Procumbent. Prostrate, creeping along the ground.

Raceme. An unbranched flower cluster with several or many stalked flowers borne singly along a main axis, the youngest at the apex.

Ray floret, ray flower. One of the flowers, usually with strap-shaped petals, that together form the outer ring of flowers in a composite *flower head* such as a daisy.

Ray petal. The petal or fused petals, often showy, of a *ray floret*.

Recurved. Curved backward.

Reflexed. Bent sharply backward.

Remontant. Flowering more than once in each season.

Revert. To return to its original state, as when a plain green leaf is produced on a variegated plant.

Rhizome. An underground, creeping stem that acts as a storage organ and bears leafy shoots.

Root. The part of a plant, normally underground, that functions as anchorage and through which water and nutrients are absorbed. An **aerial root** emerges from the stem at some distance above the soil level.

Rootball. The roots and accompanying soil or compost visible when a plant is lifted.

Rootstock. A well-rooted plant onto which a scion is grafted; see *grafting*.

Rosette. A group of leaves radiating from approximately the same point, often borne at ground level at the base of a very short stem.

Runner. A horizontally spreading, usually slender stem that forms roots at each node; often confused with *stolon*.

Scale. 1. A reduced or modified leaf. 2. Part of a conifer cone.

Scandent. See *climber*.

Scarify. To scar the coat of a seed by abrasion in order to speed water intake and hence germination.

Scion. See *grafting*.

Scree. An area composed of a deep layer of stone chippings mixed with a small amount of loam. It provides very sharp drainage for plants that resent moisture at their base.

Seedhead. Any usually dry *fruit* that contains ripe seeds.

Seed leaf (cotyledon). The first leaf, pair of leaves or occasionally group of leaves produced by a seed as it germinates. In some plants they remain below ground.

Self-seed. To produce seedlings around the parent plant.

Sepal. Part of a *calyx*, usually insignificant but sometimes showy.

Series. The name applied to a group of similar but not identical plants, usually annuals, linked by common features.

Sessile. Without a stalk.

Sheath. A cylindrical structure that surrounds or encircles, partially or fully, another plant organ such as a stem.

Shoot. The aerial part of a plant which bears leaves. A **side-shoot** arises from the side of a main shoot.

Shrub. A plant with *woody stems*, usually well-branched from or near the base.

Shy-flowering. Reluctant to flower; producing few flowers.

Simple [of leaves]. Not divided into leaflets.

Soft-stemmed. The opposite of *woody-stemmed*.

Spadix (pl. *spadices*). A *spikelike* flower cluster that is usually fleshy and bears many small flowers. Spadices are characteristic of the family *Araceae*, e.g. *Arum*.

Spathe. A large *bract*, or sometimes 2, frequently colored and showy, that surrounds a *spadix* (as in *Arum*) or an individual flower bud (as in *Narcissus*).

Sphagnum. Mosses common to bogs; their moisture-retentive character makes them ideal components of some growing media. They are used particularly for orchid cultivation.

Spike. A racemose flower cluster with several or many unstalked flowers borne along a common axis.

Spikelet. 1. The flowering unit of grasses comprising one or several flowers with basal *bracts*. 2. A small *spike*, part of a branched flower cluster.

Spore. The minute reproductive structure of flowerless plants, e.g. ferns, fungi, and mosses.

Sporangium (pl. *sporangia*). A body that produces *spores*.

Sport. A mutation, caused by an accidental or induced change in the genetic make-up of a plant, which gives rise to a shoot with different characteristics to those of the parent plant.

Spur. 1. A hollow projection from a petal, often producing *nectar*. 2. A short stem bearing a group of flower buds such as is found on fruit trees.

Spur back. To cut back side-shoots to within 2 or 3 buds of the main shoot.

Stamen. The *anther* and *filament*.

Standard. 1. A *tree* or *shrub* with a clear length of bare stem below the first branches. Certain shrubs, e.g. roses and fuchsias, may be trained to form standards. 2. One of the 3 inner and often erect *perianth segments* of the iris flower.

3. The larger, usually upright back petal of a flower in the family *Leguminosae*, e.g. *Lathyrus*.

Stapeliad. A member of the genus *Stapelia* and closely related genera of the family *Asclepiadaceae*.

Stem segment. A portion of a jointed stem between 2 *nodes*, most frequently occurring in cacti.

Sterile. Infertile, not bearing *spores*, pollen, seeds etc.

Stigma. The part of the female portion of the flower, borne at the tip of the *style*, that receives pollen.

Stipule. A small scale, or leaflike appendage, usually one of a pair, mostly borne at a *node* or below a leaf stalk.

Stock. See *rootstock*.

Stolon. A horizontally spreading or arching stem, usually above ground, which roots at its tip to produce a new plant.

Stop. To remove certain growing points of a plant so as to control growth or the size and number of flowers.

Stratify. To break the dormancy of some seeds by exposing them to a period of cold.

Style. The part of the flower on which the *stigma* is borne.

Subglobose. Almost spherical.

Subshrub. A plant that is woody at the base although the terminal shoots die back in winter.

Succulent. A plant with thick, fleshy leaves and/or stems; in this book, it is evergreen unless otherwise stated.

Sucker. A shoot that arises from below ground level, directly from the *root* or *rootstock*.

Summer-deciduous. Losing its leaves naturally in summer.

Taproot. The main, downward-growing root of a plant; it is also applied generally to any strong, downward-growing root.

Tendrill. A threadlike structure, used to provide support; see also *climber*.

Tooth. A small, marginal, often pointed lobe on a leaf, *calyx* or *corolla*.

Tepal. See *perianth segment*.

Tree. A woody plant usually having a well-defined trunk or stem with a head of branches above.

Trifoliate. With 3 leaves; loosely, with 3 leaflets; **trifoliolate**, with 3 leaflets.

True [of seedlings]. Retaining the distinctive characteristics of the parent when raised from seed.

Truss. A compact cluster of flowers, often large and showy, e.g. those of pelargoniums and rhododendrons.

Tuber. A thickened, usually underground, storage organ derived from a stem or root.

Tubercle. A small, rounded protuberance; see also *areole*.

Turion. 1. A bud on a *rhizome*. 2. A fleshy, overwintering bud found on certain water plants.

Umbel. A usually flat-topped or rounded flower cluster in which the individual flower stalks arise from a central point. In a compound umbel each primary stalk ends in an umbel.

Upright [of habit]. With vertical or semi-vertical main branches.

Water bud. See *turion*.

Whorl. The arrangement of 3 or more organs arising from the same point.

Winged [of seeds or fruits]. Having a marginal flange or membrane.

Woody-stemmed. With a stem composed of woody fibers and therefore persistent, as opposed to soft-stemmed and *herbaceous*.

A **semiwoody stem** contains some softer tissue and may be only partially persistent.

× The sign used to denote a hybrid plant derived from the crossing of 2 or more botanically distinct plants.

+ The sign used to denote a graft hybrid; see *grafting*.